



CEBU INSTITUTE OF TECHNOLOGY
U N I V E R S I T Y

IT342-Section SYSTEMS INTEGRATION AND ARCHITECTURE 1

FUNCTIONAL REQUIREMENTS SPECIFICATION (FRS)

Project Title: **Mini App – User Registration & Authentication**

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1. Introduction

1.1. Purpose

This document provides a detailed overview of the User Authentication and Profile Management System. It serves as a blueprint for stakeholders to understand functional and non-functional requirements, with emphasis on secure access and data persistence

1.2. Scope

The system provides a secure gateway for user access to protected features. It includes:

- User registration
- Secure login
- Profile/dashboard access
- Session termination (logout)

1.3. Definitions, Acronyms, and Abbreviations

API (Application Programming Interface): Rules enabling the React frontend to communicate with the Spring Boot backend.

JWT (JSON Web Token): A compact token used for secure authentication and authorization.

2. Overall Description

2.1. System Perspective

The system follows a client-server architecture where the React UI handles user interaction, and the Spring Boot API manages business logic and database communication.

2.2. User Classes and Characteristics

- **Guest User:** Unauthenticated users who can access landing, registration, and login pages only.
- **Authenticated User:** Users with a valid token who can access the dashboard and profile data.

2.3. Operating Environment

- **Frontend:** React.js (Node.js environment), modern web browsers
- **Backend:** Spring Boot (Java JDK 17+)
- **Database:** PostgreSQL or MySQL
- **Tools:** VS Code / IntelliJ IDEA, Postman for API testing

2.4. Assumptions and Dependencies

- **Assumption:** Users have a stable internet connection to communicate with the API.
- **Dependency:** The system depends on the availability of the Database server for user credential verification.

3. System Features and Functional Requirements

Describe each major feature of the system and its functional requirements.

3.1. Feature 1:

Feature 1: User Identity Management (Registration and Login)

Description: Entry point for users to create and verify their identity.

Functional Requirements:

- **Account Creation:** Allow registration with unique username, email, and password.
- **Credential Hashing:** Hash passwords before storing in the database.
- **Identity Verification:** Grant access only if credentials match stored records.

3.2. Feature 2:

Feature 2: Access Control & Profile Management

Description: Manages session and protects private pages.

Functional Requirements:

- Session Authorization: **Issue a secure JWT on login for dashboard access.**
- Page Protection: **Prevent unauthenticated access to protected profile pages.**
- Secure Logout: **Clear the authentication token on logout.**

4. Non-Functional Requirements

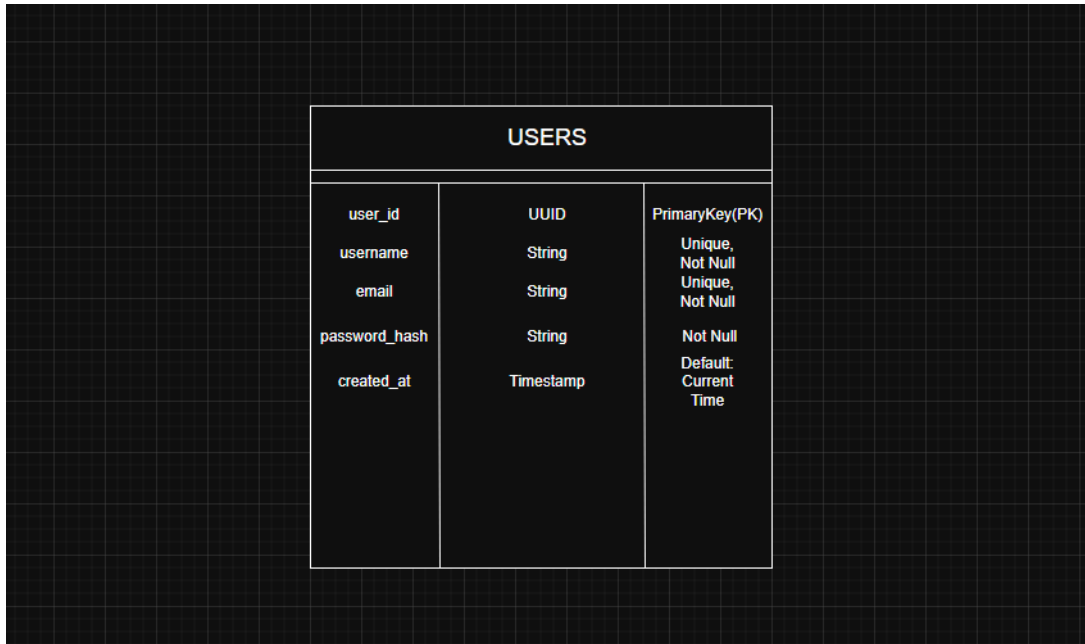
Attribute	Requirement
Security	Protects against unauthorized access by using industry-standard hashing and token-based authentication.
Performance	Authentication and profile loading should occur within 500ms to ensure a responsive user experience.
Usability	The interface should utilize a clean layout with consistent typography like Montserrat or Playfair Display
Reliability	The database and API must maintain high availability to prevent users from being locked out.

5. System Models (Diagrams)

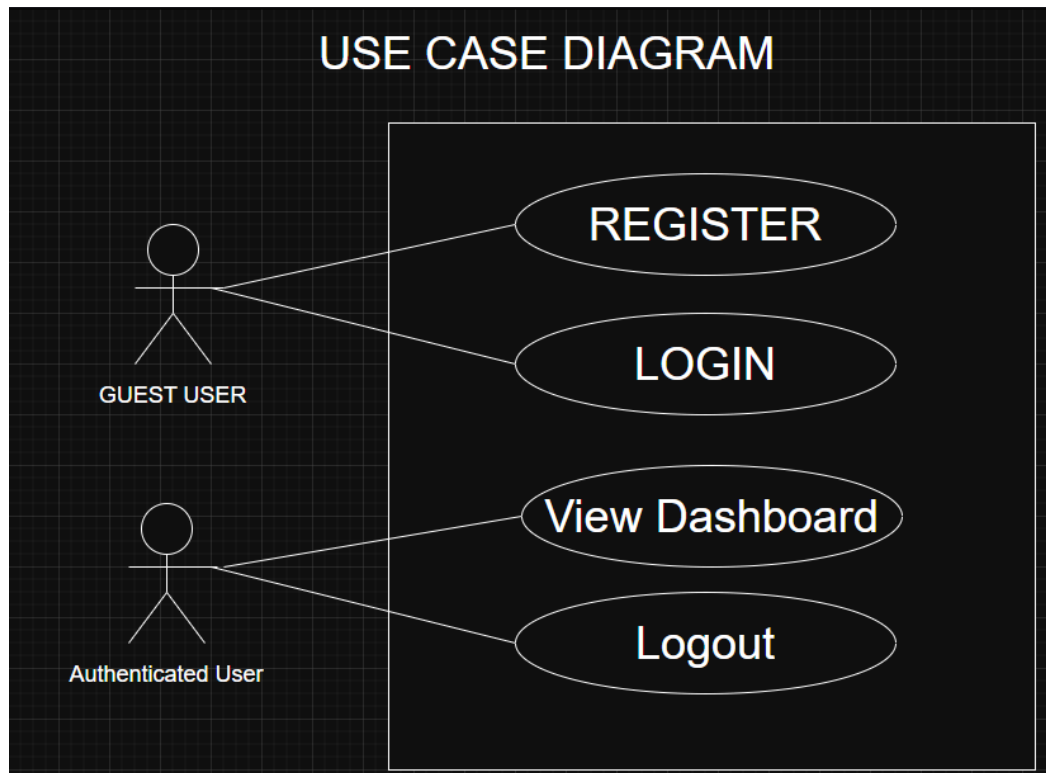
Insert the necessary diagrams for the system:

5.1. ERD

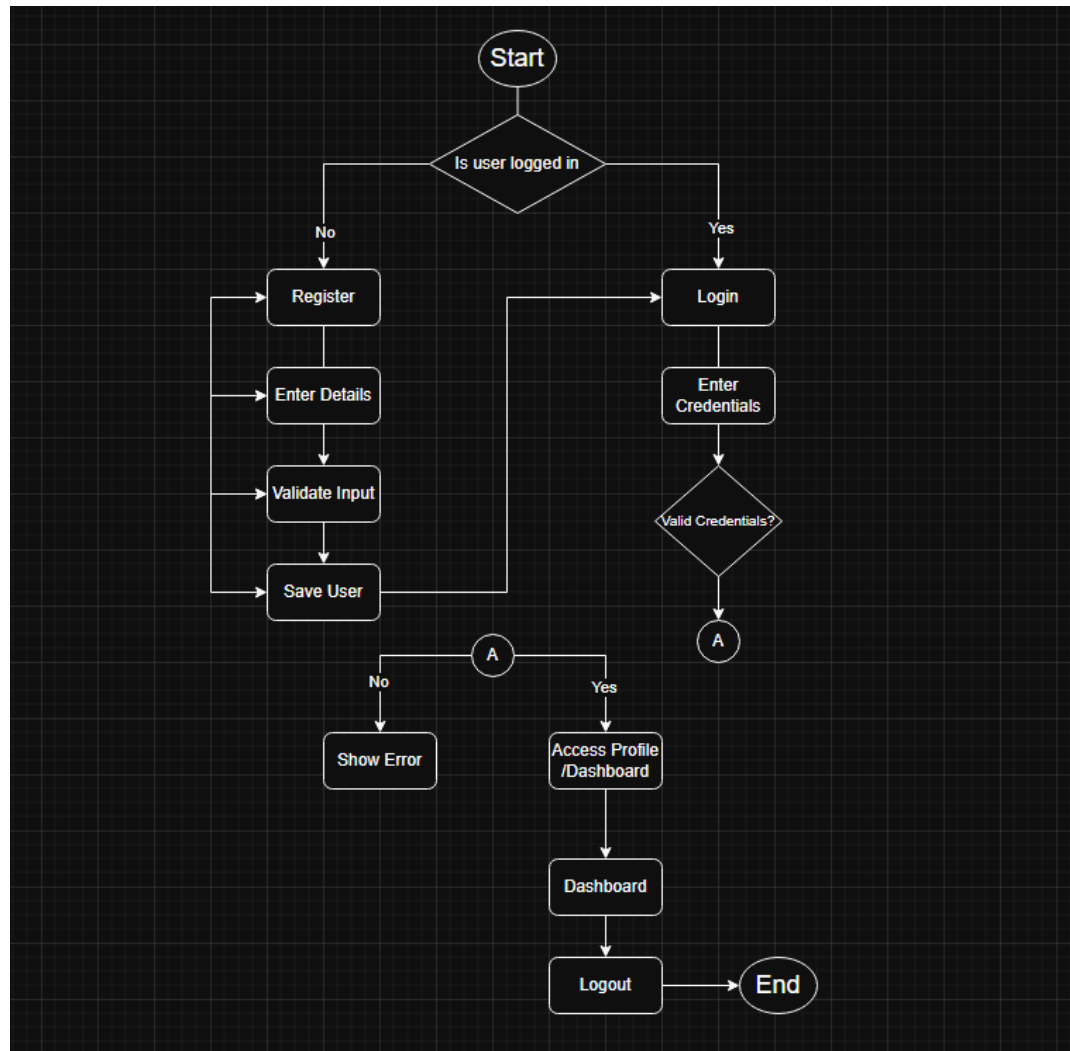
ERD



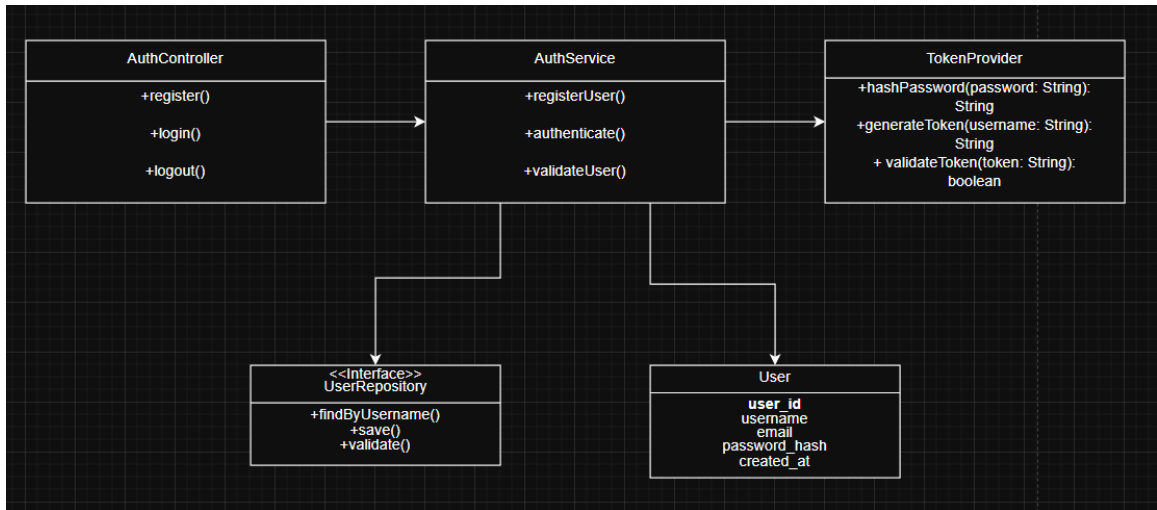
5.2. Use Case Diagram



5.3. Activity Diagram

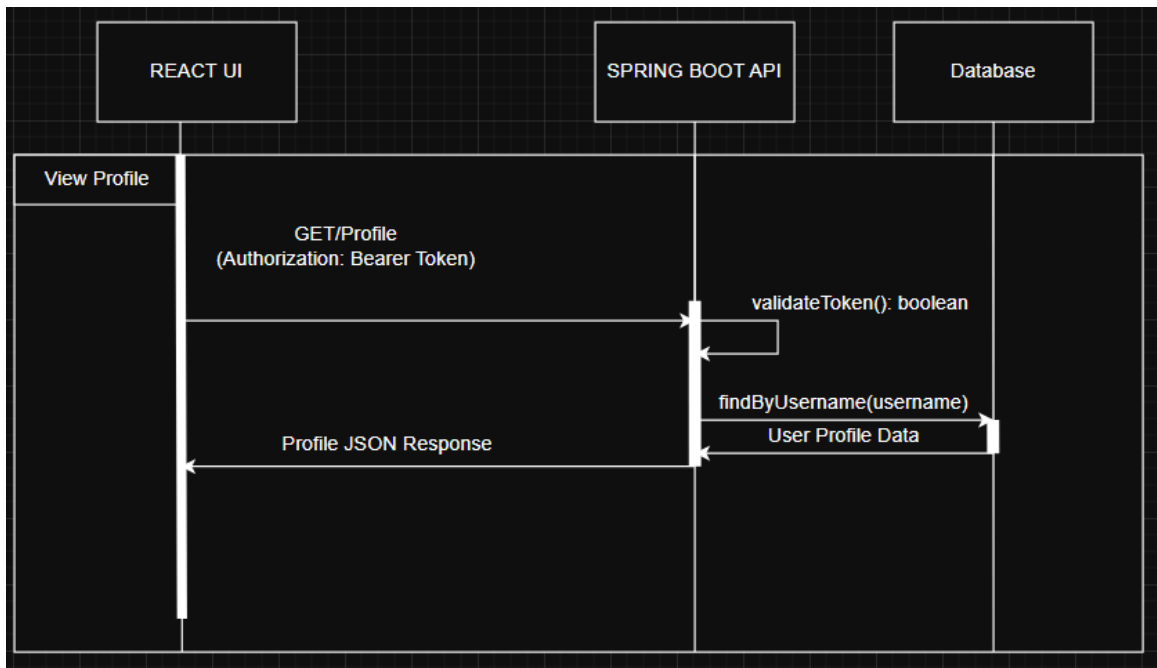


5.4. Class Diagram



5.5. Sequence Diagram

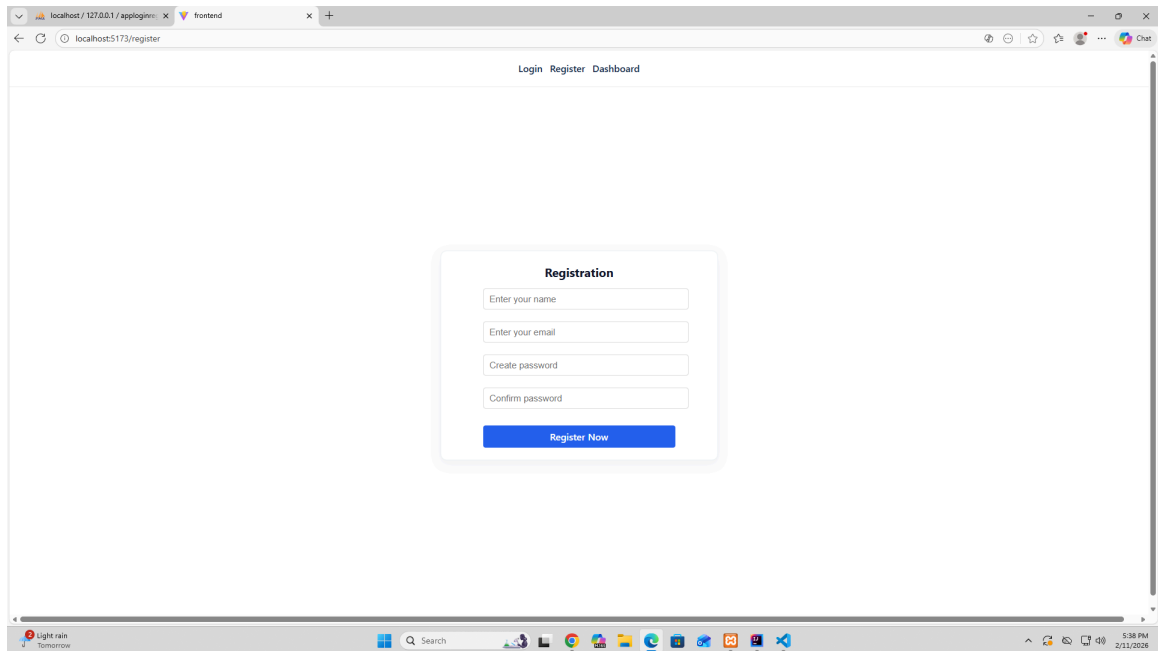
SEQUENCE DIAGRAM



6. Appendices

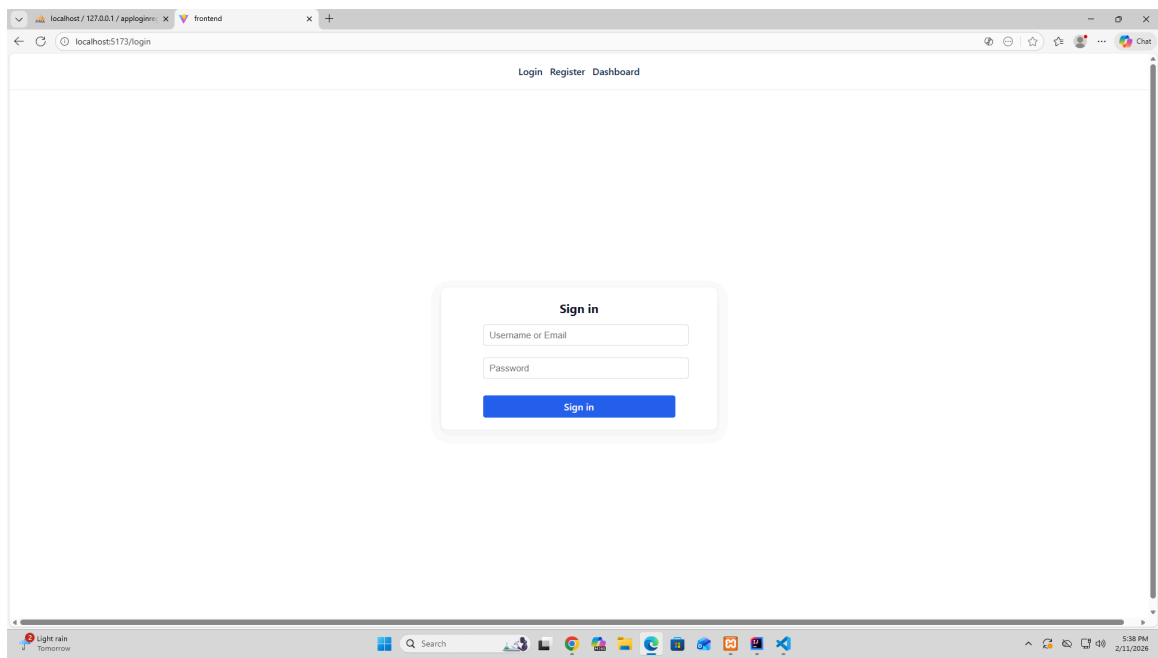
Include any additional information, references, or support materials.

Register



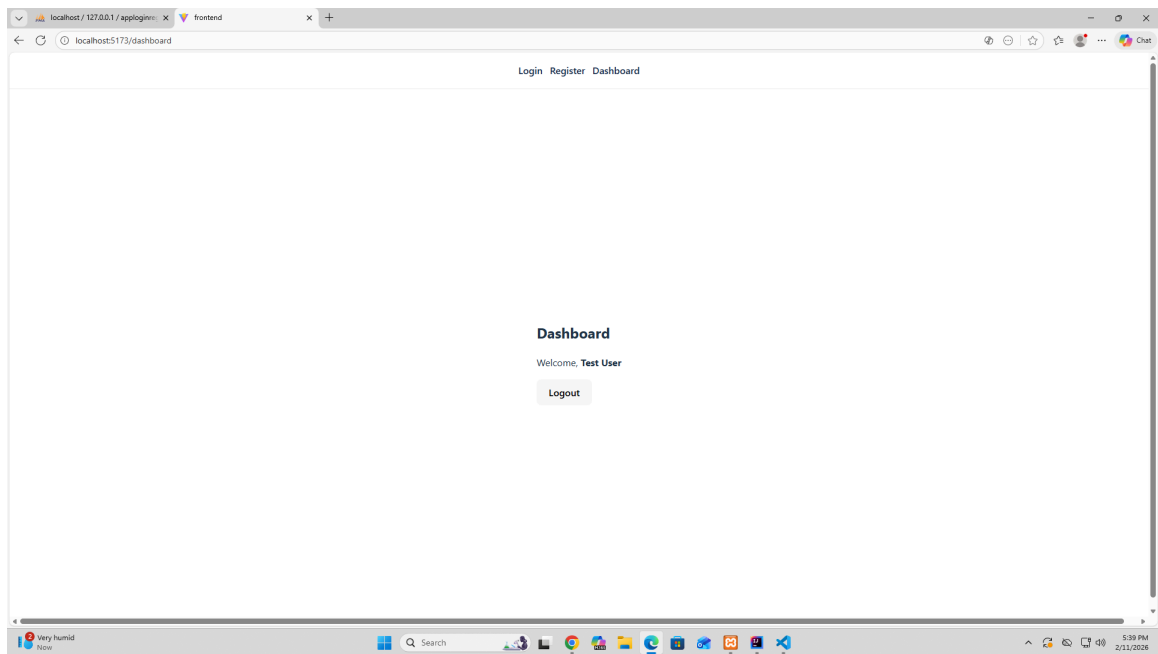
The screenshot shows a web browser window with the URL `localhost:5173/register`. The page has a navigation bar with links for `Login`, `Register`, and `Dashboard`. The main content area features a registration form titled "Registration". The form includes four input fields: "Enter your name", "Enter your email", "Create password", and "Confirm password". Below these fields is a blue button labeled "Register Now". The browser's taskbar at the bottom shows the Windows logo, a search bar, and various application icons. The system tray indicates "Light rain Tomorrow" and the time "5:38 PM 2/11/2026".

Login



The screenshot shows a web browser window with the URL `localhost:5173/login`. The page has a navigation bar with links for `Login`, `Register`, and `Dashboard`. The main content area features a login form titled "Sign in". The form includes two input fields: "Username or Email" and "Password". Below these fields is a blue button labeled "Sign in". The browser's taskbar at the bottom shows the Windows logo, a search bar, and various application icons. The system tray indicates "Light rain Tomorrow" and the time "5:38 PM 2/11/2026".

Dashboard/Profile



Logout result (e.g., redirected to login)

